

ActInf GuestStream #028.1 ~ Reconceiving rationality: situating rationality into enactive cognition

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hello and welcome everyone this is active inference guest stream 28.1 it's October 14 2022. today we are going to be hearing and discussing reconceiving rationality by Giovanni Rolla so Giovanni thank you so much for joining very much looking forward to this presentation and discussion so off to you thank you thank you then it was a pleasure to to receive your invitation a couple of months ago and I am delighted to be here to have this opportunity to discuss my work um I'll spare your request um our suggestion I am uh presenting this paper I wrote a couple of years ago and uh as we were talking the backstage I have I've changed my mind a few times and I'm going to make this clear during the presentation since I published that paper but it is a pleasure nonetheless to have the chance to present this video here and uh to to share it with the world so I'm very happy uh so we'll be talking about reconceiving rationality and this is the this is the the itinerary I was thinking about presenting to you so first I'll be talking about rationality and contentless cognition then I'll present my view which I called radically inactive rationality and safety words about agency and proficient interaction and reasoning and rationality they are close with a couple of objections the first one being that whatever I'm pressing here is uh too liberal in a way excessively liberal it encompasses too much and the second objection which I think is the one that may interest to you the most is whether it is in fact compatible with predictive processing or predictive coding which is of course in the facility in the end it's near the the tour of cars in central to do to active Infinity so here's the here's the sorry the my initial thoughts on the matter is that traditionally the very uh Orthodox have view on rationality uh assumes to most of my knowledge it's as an assumption really that rationality must entail reason that for instance I believe B in an action a are rational if and only if there is the reasoning that underwrites in the Bure so and of course it's just not any reasoning the reasoning question must acquired with certain certain formal benchmarks such as classical logic or probability reasoning or heuristics so believing and acting rationally

according to this traditional field require meeting certain formal standards and this as I said it seems to me to be largely an assumption about what it means to be rational about defining rationality in terms of reasoning so if rationality requires reasoning and if reasoning involves the evaluation of beliefs against certain formal benchmarks then rationality implies content for cognition where content I take it to be minimally having accuracy conditions being capable of being right or wrong or in or in on a more weightier notion being true or false so the idea here is that rationality necessarily involves content it involves beliefs reasoning and the comparison or sorry to update right now one software update I'm sorry there was a pop-up in my window over here so um that would be I think the the standard view on rationality and if it is so that rationality necessarily involves content it it is a good fit with classical cognitivism or cognitivism in general which is the view that combination necessarily involves representational content at the very least for some cognitivists of course it might involve also conceptual content but at the very least representational content which has uh accuracy conditions but this classical view is not so good it does not provide a so good fit with radically inactive and embodied cognition which I am here uh thinking not specifically on radical and activism by Goodwin Maine in the recent books but I think it also challenges other views that are not presented as radically and active but of course they share some core tenants with radical and activism we could call them uh radically embodied views of cognitions such as um with the views presented by Ezekiel Paulo sensory model life you also have Gallagher's enact variety variety of inactivism in his 2017 book and also the I mean the view by by Tony kimero in his 2009 book on ecological psychology and dynamical systems they all share common tenancy and here I am I know this is very controversial but I'm treating those views as radically embodied views of cognition and but I'm specifically focused on radical and active views of cooking issue they all say that uh not the recognition is representational contentful which is in a way to say that we can think of basic contentless forms of cognition or a contentless level basic level of recognition which emerges through the active exploration of one's environment and Beyond radical and activism by what we mean this could be this idea could be cashed out in terms of mastering sensory motor engagements a matter of know-how the knowing how to engage with your surroundings instead of knowing that stuff are so and so around you so this is very clearly what the Paulo and colleagues and uh kimero for instance do in their books uh radical and activism has a certain tendency to skew sensory model engagement sensory model schemes Etc and uh but recent developments have I think Incorporated that so the the this is the use of the the list of this idea right so if we combine

radical embodied views of Illumination or right connectivism and the classical fuel on rationality what we have is something that angle has described as follows and activism is the view that cognition is grounded in a pre-rational understanding of the world that is based on sensory motor acquisition of real life situation and he explains these uh this idea of pre-rational understanding by by the the ideas of Merleau-Ponty Heidegger according to whom the relation to the world can be only one written in practice inactives in acting and practicing intern is mediated through the body so what seems to be implicit here is that unlike acting and practice rationality is not actually it is something that is not at a very basic level of combination so this is what happened when we we make room for a basic level of cognition rooted in active and in practice and divorce it from upper levels of combination that they are usually the locus the lucky Clarity of rationality so the idea in summation is that contentious cognition or basic cognition that which emerges through sensory motor engagements is not rational so sensory modern engagements however virtues in prolific they can be cannot be evaluated from the standpoint of rationality only content for cognition high level combination can be rationality is expressed for instance in beliefs and reasoning and well this is something that I haven't touched upon on the paper but it was my implicit motivation to deal with this subject as someone that comes from philosophy of the mind is the idea of the myth of the given at least a version of it it seemed to me at the time and uh I was correctly persuaded by a reviewer not to touch on the subject it was very convoluted in my initial presentation but it seems to me that if we divide base recognition from a rational thought and action and we do not Point towards the well there is no I mean intersection between these two levels it seems that radical and activism and uh similar views would be committed to the myth of the given unpopularized by Sellars and moderations by McDowell as a problem for imperious fields of my mind so it seems that this is I think a [Music] a philosophically interesting problem but it's not something that is uh on the Forefront of the paper it was my implicit motivation here that to explain how can we combine the contentless or basic level of cognition through sensory modern engagements with rationality if you cannot it seems that uh there is a given of some sort that is being uh not completely explained in our picture of the mind so this is something I wanted to I still want to try to explore this this intuition more clearly but I haven't touched upon this but I I think there's something in the direction of what why what I will show today is I think it's a way to tweeze this this accusation of being committed to the myth of the given so how can we reconceive rationality I mean I think in order to avoid that issue and avoid the idea that we have two levels completely separated from each other where one The Higher One

is the locus classical rationality and what else is beneath it is not rational I think the idea is to rejected rationality must be defined or it entails necessarily reasoning and nationality does not need to be still conceived but then the challenge becomes to offer a positive proposal of what rationality is if not reasoning and here is Susan Hurley in a paper that my knowledge is the only one that it's upon this subject and I think she was also concerned about the myth of the given by the way at the time and here is a long quote from her phenomide me reading it is that rationality might emerge from a complex system of the centralized hierarchy relations of inhibition facilitation and coordination among different horizontal layers Each of which is dynamic and environmentally situated rationality reconceived in horizontal or modular terms uh I don't know if she has to use this terminology but in any way that's how she chose to express it is substantively related to the environment it does not depend only on internal procedures that mediate between input and output such as content either for the organism as a whole or for a vertically bounded Central cognitive module rather it depends on complex relationships between dedicated World involving layers that monitor and respond to specific aspects of the natural and social environment and of the neural network enriches the feedback from their responses this to my knowledge is the earliest case of someone working on the inactive approach in a way is obviously more usually associated with the work of avano which is also an activist but in a different approach from radical and activism and here is the the earliest and to my knowledge before I wrote that paper is that the clearest uh proposal of rationality reconceived in the native terms but it's not it's not something that uh gained traction after her publication to my knowledge actually I haven't seen much since then and to summarize the idea is that we have to change the explanatory Focus which is rationality is not exclusively thought of as a Content involving capacity in this proposal it is situated or World involved in increased words and is it is radically embodied it depends on bodily morphology and action and how the the organism is situated and acts in that environment the idea will be minimally dead so this is what I call radically inactive rationality is uh in a very uh loose definition so far is that an Asian acts rationally in so far as she maintains a proficient interaction with her environment coordinating her green capabilities according to environmental conditions of constraints and solid stations and of course this sometimes requires adapting to unforeseen circumstances the the the core intuition is this and it's interesting to compare this idea of radical and active rationality with another of Susan Hurley's ideas in a text from 2003 she's she says reasons uh for acting she argues that reasons for acting maybe context bound where she's having against uh Gareth Evans generality constraint and

this is the defense of the idea of islands of practical rationality so she is still treating rationality as essentially something that relates to reasoning here but this kind of reasoning is uh bounded by practical context so this is a very Progressive view if compared with the traditional view I I talked about at the beginning but still more conservative than what I am trying to develop here so another thing that I think must be uh the Forefront of this proposal is that rationality implies agency and autonomy because something or only something that an organism does can be said to be rational right so to use these uh the ideas from the autonomous and activists camp and from the theory of how to players is autonomous systems such as a Roomba cannot be rational because it's not an agent it is not the author of its own behavior it is designed by external agents to work in a certain way so it's not strictly speaking Nation so that is where I would draw the line of where well you cannot it makes no sense to ascribe rationality to a Roomba for instance for this I I use the the Notions of autonomy and adaptivity the DiPaolo and others you have also the the recent book image the four sensory model life and that famous paper from the bowl from 2005 where he talks about the adaptivity to to set his very general constraints on what it means to be Russian so autonomous systems are thereby conceived as precarious networks a precarious network of processes whose organization and physical boundaries are produced by the system itself the system is continually maintaining its own uh viability conditions which is uh known as sense making where the system explored uh environmental environmental conditions that are favorable to its uh self-maintenance and avoids the ones that are detrimental to it so this is how I understand to be the the very in general that the emotional constraint on a rational being that it must be an autonomous system and the interesting thing here is that the concept of adaptivity that Paulo puts forth is that the capacity of an organism to improve its condition of self-production through internal structural changes so it involves the monitoring and regulation of internal changes in order to select the best available course of action it is I think it's another interesting concept to think in of to to include in this idea of a radical and active notion of rationality one that is not necessarily committed to reasoning and content so agency is realized through the exercise of situated abilities that promote changes within the organism in accordance with environmental constraints and solid stations which means that how well an organism adapts in order to deal proficiently with its environment is a measure of or an indication of its rationality the more rational an organism is the more suited it is to deal with its environment and we have foreseen circumstances and in doing so the organism enhances The Continuous exchanges between itself and the environment which is crucial for its survival and it is very naturalized way of

thinking of rationality is mostly capacity to to enhance changes for survival this is the the overall idea here so as I said before what about reasoning I want to I don't want to commit rationality necessarily with reasoning and this is what radically inactive rationality is all about but I don't want to exclude reasoning as a former professional combination so what I'm saying is that it's not defined by reasoning but it's not uh it does not exclude reasoning reasoning I think would be thought of as is special or specific mode of rational agency one that is exclusive to our knowledge to human beings but it's not the defining feature of rationality so for this I uh in order to to explain this idea I recall how Hutu and Saturn in between their people in a most recent book by what I mean they explain the emergence of high recognition the idea is that being able to engage in content for cognition ultimately depends on being social culturally embedded where content full cognition emerge from social cultural interaction so as an individual takes part in Social cultural practices they accept more ancient cognitive abilities to deal with symbols and by doing so they um in a long process of enculturation of course they become able to engage with a symbolic coordination which is not internal representation specific because it is publicly available it's not mental representations in the traditional sense but it has the features of symbolic cognition without the commitment to there are no representation or mental representations so acting rationally in its most General sense is the idea that the proficient behavior of an organism is in is the proficient behavior of an organism in its environment it depends on the coordination of embodied abilities whereas reasoning if everything goes well of course is understood as content for rationality for my specific mode of rationality reasoning is the proficient behavior of an organism in a social cultural environment which is a more specific aspect of our environment of course and it depends on the coordination of such a culturally based or content involving abilities which are more recent in the the human phylogenetic scale so well this is the the biggest of the idea this is the the I think uh the the positive part and here are the objections I wanted to talk about first one being that it's a very liberal view it's very uh it it is well it decompasses too much one would say because if rational action boils down to the coordination of cognitive abilities then rationality is something gradual for an organism could be more or less proficient in its environmental interactions and the the rejection goes according to radically inactive rationality many organisms achieve and maintain an Adaptive behavior in the niche with different levels of success to which I am uh well someone could sing rational and active radical and active rationality is to Liberal and this is one objection that I've heard in person and I I kept it so hard because I think it's very it's a very good way of putting it even snails are rational on this View

and uh to which I must say that I'm willing to to buy the bullet uh well yes it's more liberal than traditional views of course it is intended to be so but I don't think that this per se problem what I think is the the positive part here is that it preserves a difference of degree humans are way more rational than other animals simply because we are able to perform a wide variety of actions in a wide variety of dynamic circumstances that are Way Beyond any other animals that we are aware of so to come back to the example of this nail this nail can perform a few complex actions can move from one point to another it can search for food and shelter and so on so forth but they have a critical limited set of abilities if you surround this nail with South for instance it is doomed to to die by the retortation and humans on the other hand they are much more adaptive and able to deal dynamically with many unforeseen circumstances so uh in my response that the rejection is that yes uh well it is very liberal but that's not the problem the problem would be if that it is if it had the room for a difference of degree between how animals deal with their environments so and and this is the other object objection which I think it may be though the one that interests you the most then you know is the 12 predictive processing seems to imply inferences and content all the way uh but at first I must uh warn you and the viewers that at the time I wrote this uh this paper was much more optimistic about this possibility of combining an activism and predictive coding or predictive processing and uh I think there are important points here but there is this recent paper by Olu and Evan Thompson and Randy beer where they uh they provide a critical assessment of uh the the the attempts to combine mainly the free energy principle and inactive and autobiographic theories so I am now mostly convinced by their arguments but I think there are some interesting points of comparison here nonetheless so predictive process in our predictive code in uh at the time I was writing in still to this day is all the raged in the cognitive Sciences I guess you can say that it sounded sweets become the one of the main paradigms out there so it is interesting to see if it provides a good fit with what I'm talking about and in these viewers you know competition is understood as a process of error minimization that occurs in bidirectional narrow hierarchies you have the generation of internal models for the activity of lower layers that anticipates the distal causes of sinisterial stimuli which is a top-down process and the minimization of discrepancy between prediction and current stimulus signal by um the operation of priors which is a bottom-up process so here's of course active inference which is understood by for instance in an informal you know formal terms by Clark an organism moves its sensors in ways that amount to actively seek in our generation generating the sensory consequences that they are rather their brains expect this is one way of understanding active

influencing very among formal terms but I think it gets the job done so is the brain inferring the expected source of received stimuli if it is it seems that there is a there's an issue with what I'm talking about so the Hamilton reading of predictive processing which is uh which uses the broader bias and inference right says for instance that the process of perception is separable from rational provided by agents process of proliferation and context top-down effects I felt that every intermediate level of processities has caught my attention the way that Clark is also talking about rationality all the way but it's an inferential sort of and pervasive influencer reasoning seems to imply impervasive content and then predictive processing seems to be incompatible with radically inactive well but there is uh the time there was and I think we can still argue in several of the there's no non-hemorrhoids and reading of predictive processing which uses for instance a one first step would be to use a deflationary sense of prediction that understand Camaro in a short paper from 2013 but fourth where the values of two or more variables cover your value reliably so the value of 1 allows the Observer to infer the value of another and here the inference is happening happening at the level of the observation not at one far variable is inferring the value of another so the idea would be I think that content is in the eye of the beholder the person who interprets how the selection of censorial we put approximates internal predictions he understood in the deflationary sense can describe it in inferential and biasing terms so this is how I see for instance the the idea about fourth house by Bruno marstein in that paper which if I'm not mistaken title is something like the brain is not a scientist and the argument is in this General interaction that we are projecting how scientists construe brain activity as a feature of the brain itself as if it were predicting an inferring this more uh robust or waiter sense so what I'm trying to say here is that active inference then I use double I want to say double forgot the name of this in English sorry um it's not uh the literal sense of inference here according to radically inactive rationalities that unarmed for the efficiency of co-oriented behavior the agent has to act in certain ways in order for his his or her States come to fruition more efficiently less costly this does not imply the more contentious claim that the generative models themselves display and perform predictive inferences this is a very deflationary reading into predictive processing to to provide this combination with radical and active and here is a kickoff in a paper from by model it doesn't follow that an organism has an internal representational model of its niche in that it is this model that those are the cognitive work if you like instead an organism is a model that is the causal and statistical regularities reflect in some environmental reflecting some phenotype there is a model and well this relates to something that recently uh my colleague notifica and I

have been talking about the idea of how organisms bring forth a world according to an activism which is idea that the organism and environment are historically historically called determined so uh artificial genetical scale this could be explained through the idea of Niche construction the idea that once an organism is active in its environment it promotes changes then if they are stable enough that it can be inherited by its Offspring and these changes provide new evolutionary press pressure pressures that are uh that lead its evolutionary path a similar idea has also been explored by other description this time relating Niche Construction and the ecological psychology so my my response to that objection is that to act rationally means to achieve a proficient engagement with the environment through in a way prediction and urbanization I'm willing to accept that characterization but rational and broader biasian processes need not imply content for cognition if we if we understand that in the Flash no sense prediction and narrow minimization involve the selection of input and the coordination of an interabilities in World involvement manners which is well exactly the main claim of radical and active personality so I think to that extent that predictive processing is not uh an enemies would be rather than a lie of this idea instead of uh arrival but of course there are other issues of compatibility that I haven't I haven't touched upon here but that uh this is all that I have thought of but today thank you awesome thank you very much all right cool very interesting for those who are watching live please feel free to write any questions I will gather my notes and let's begin the discussion okay well first just uh thanks again for the talk how about a little context on how you came to be asking these questions in your current position and how your interests in this area developed because there's so many interesting ideas and I just wanted to learn more about how you came to this Framing and sharpened your point okay thank you that's that's a good question because it's not often that you can explore this background it matters so uh I was at the time I think I was writing my PhD dissertation it was maybe 2007 or something and uh I was arguing for the idea that rationality must be something that is ultimately World involved in there is a tendency to think of well even among as I told as I showed you among uh in in embodied periods with people of certain it needs to be essentially World involving such as acting and practice and so forth and perception of course but rationality remains something like not realizable notion something that cannot be brought down to the level of the the it could not be scaled down in a way like perception that could be explained by a certain Dynamic engagement and so on and so forth so at the time I was thinking that well maybe this is an issue because on other other contexts it is taken to imply that whatever perception moves if it is not articulated in a way to provide a certain fit with

professional combination then perception is a given and this is the myth of the given I mentioned at the beginning and I thought that well maybe this is something of there is something of a myth of the given working here on an active cognition but this is not I think it's not fatal because we couldn't principle argue that instead of uh construing step perception and other basic forms of combinations involving Concepts and the like maybe we could argue from the other direction try to scale down rationality in more not naturalized level of the perceptual organization and the like so that was the motivation and it was very it was a long review process and I believe that my first the first version of the paper the the viewer was very I think they were correct entirely I I endorsed the suggestion after all but he was very uh they were very adamant on taking that apart and I think it was for the better because it it is another literature it is another uh a lot of references that are not usually in conversation with an activism and body recognition and it was very difficult just to put the matter in the same terms to advance my arguments so it's something that I scratched out and I haven't thought too much about that system but I think it's something I would like to come back to and maybe uh even probably someone has done so in the in this last few years I'm not aware I can remember but this relation between an activism the myth that we give and it seemed to be it seems to be very obvious so it is I think something that we could we should I mean uh be able to explore but there's not something that was in the the Forefront of the paper but but I had this idea to talk about rationality in the flashing every sense and at the same time I was reading a lot about uh recording from a philosophical perspective of course I'm not as I said very well versed in the technicalities and the formalities of the the approach but I was interested in the the philosophical consequences there and I thought well maybe there's something to compare here because as Clark's quotation show you can talk about rationality all the way and uh that's something I wanted to to develop but not in a cognitivist view but not radically embodied film so that's the the the background awesome thank you could you maybe describe just what is the myth of the given what was the context that that myth arose in and where do we go after bringing up the myth of the given because I I saw it as a concept that you wove in so what was it about okay so the middle beginning that well there are several Givens depending on your flavor of philosopher's mind but most classically in previous fields of the Mind take the the the content of perception to be given prayer rationally in a way that we have certain inputs that are uh just received by the mind and the mind interprets it so sellers different sellers is famous for uh having described his uh this problem as a problem for not only increases fields of the mind but also for rationalists but on the on the other side rational is usually taken uh rational

intuition as a given which is to say that there is some some level of cognition that is uh unarticulated it's given or as some people talk about being bred given but in a way there is this level of cognition that is given in the mind has to interpret it but if it is given in this way it has no point of contact no uh intersection between whatever whatever mind does and the in this basic level of uh domination so my idea is to avoid the the charge of being committed to the myth of the given but again this is not something that is explicit in the the next literature but it's something that seems to be possible to to develop in a I mean uh in this sort of comparison but it's not it's not from the not clearly there but one way of solving them for the given is to say that perceptual content is already conceptual in nature which is for instance how Joe McDowell does in mind the more of that in a way perception uh the the joke is comes with subtitles you interpret the world as as you do but uh I think that one way or another way is to demystify you know a rationality and the upper layers of combination try and try to think of them uh in less uh content for analysis without committing them so it's strictly speaking with a level of content that is divorced from basic levels of domination that's one way I think of doing so and rationality seems to be the one key concept to do this because it's so usually soon to be something exclusively human and something that is related to reasoning and well if it is so then rationality cannot inform action and if of course if action is contentless in nature so if rationality is a sense essentially contentful and it actually is essentially uncontent for contentless then you can you do not have this uh that's done this uh intersection between two domains and this is something I wanted to avoid but as I told you it became so hard to express this views in in sufficiently clear and general way is that by recommendation of a reviewer I scratched it out and I I still think about it but not in I haven't developed a proper line of argumentation that deals with it more clearly than I can do as I just did in very general terms cool well some words that you associated with rationality were adaptivity and also proficiency so what do you see as the relationship of rationality and Fitness as a biological function right yes yeah okay I um I think it is it is an intrinsic Revelation really I think it is uh probably the most uh um obvious consequence of rational action at a special scale and when we talk about whether rationality is a feature of uh certain species or instead of uh individuals I think Fitness is one of the the key concept here to to think this as the the outcome of this rational action so although I do not say I do not put eating those matters in the paper I think it is possible to think it that way and don't see it as a uh as an objection to think think of rationality but again it's very liberal and I'm willing to to accept this consequence that well every living organism in a way if it if it hasn't gone Instinct yet so it is in a way fitted to its environment so I'm willing

to say that it may involve a minimal level of rationality of dealing with uh the negatively changing environment but well it's the price of having a very uh for labor or Progressive notion of rationalities this is a consequence of it I believe so you've said that Fitness is a consequence of rational Behavior are there other consequences or how do we recognize rationality from the outside other than just saying well the ant colony is thriving so to the extent that it is rational it is adaptive an exhibiting agency and all of that um is this such a deflationary rationality concept that all extant things humans and otherwise are merely described by it or are there some outcomes that we can use to disentangle like for example a rational or irrational behavior from merely the consequences reflected by survival and right well I I had to think very carefully about this because I don't want to trivialize this concept in a way that well everything is IPS of fact or rational by the matter of fact that it's living it is acting relationship but I think that uh in particular cases one could uh identify the nation that I am equal to the individuals um self-sustainment in that Niche so so at the individual level in particular okay this is one conclusion you see certain actions that could classifiers irrational to the extent that they uh are detrimental to to with survival but uh at the species level it's more difficult to argue that way because to identify this kind of behavior or consequence but I think that in order to not to trivialize the idea of rationality and say well everything that is living is it's a factorized you'd have to to identify this sort of particular acts that are detrimental to to one survival so in the case of the the ant colony I don't know maybe welcoming invaders would be a would be the sort of behavior that would be prejudicial with survival as such I don't know I don't know if this is uh it's one impossible way of thinking about this I had to think more clearly cool very interesting and you brought in in that answer and in the presentation like multiple nested time scales just like Eco Evo Devo does from a more biological perspective looking at how Niche construction Bridges the gap between slower phylogenetic processes and more rapid developmental processes um so how does this embedded and multi-scale consideration relate to the the content concept like is content going to be localized in one body or is the content distributed across entities or is the rationality distributed across entities how would we even identify the locus of rationality and that also kind of invokes the um horizontal and modular conception from the quote that you provided so how do we identify a salient spatial temporal scales and systems of Interest oh I I don't know I don't know if I can answer uh this fully but it seems to me that at the level of uh if we're talking about content for instance content as a accuracy conditions for symbolic thoughts I think fortunately we have to say that it's it's not localized in a single individual it's publicly

shared so you can say that one consequence of this is that it's possible to think of different uh patterns of reasoning that have being information sites at different times Etc and you can see that well they developed different level different uh rules for Content information and I think this is plausible to say that content is is localized that the the uh how can you say it's share the amount you know the specifics of just for rationality I think one could identify in particular behaviors if it if it feeds these uh this kind of description of uh spin professional proficient interaction and adaptive interaction between an organism and its environment its Niche so probably we could we could give different answers here a rationality and content I I am willing to say that content for organization is not something that happens in the head something that is out there in a way shared among others and this is something that the philosophy of mine has at least since the 60s uh the philosophers of mine have been argued for the idea that content is externally individuated and uh I'm I think this is it is one of the video by for instance Hillary Putnam and Tyler Burge and others that mental content is individuated at the community level or societal level but along the on the other hand they think rationality could be in principle individuated in particular behaviors but it would depend on the individuals goals and the the niche it inhabits and so on so forth uh there are of course the actual variables here that might be important too to identify Behavior as rational or not thanks um well one phrase that really stuck with me was like know how not know that perhaps uh a basal point in your work and a focus on that action so I just wanted to hear about how action and practice as you wove them here today how are those terms addressed and how are things different today with the treatment of action and practice and know-how not know that than say in the helmholtzee in days or even in ancient or in World Knowledge traditions So Okay cool so the idea I think would be that there is a tendency from uh it is a tendency from a very modern philosophy but it has been very influential in that the first wave of cognitive science to think of perception as a capacity to categorize the environment and to offer a discrete description of environmental aspects so for instance perceiving that uh the the the the I don't know the the curve is to my left is a matter of knowing that there is a thing with searches such properties to my left and this is a kind of uh know that a kind of propositional attitude towards the world so uh I think the the interesting thing about an activism and ecological psychology as well and to the extent that they recall American pragmatism from the the beginning of the centuries to think of perception as a matter of not categorizing these properties in the world but as uh disclosing possibilities for acting and this is something that if you were to explain that in terms of uh in epistemic terms such as knowledge we wouldn't put it as a matter of

knowing that the the curve is to my left but knowing how to grab the cup and and do whatever I want to do with it if I want to drink a cup of coffee or if I want to use it as a paperweight it will have different silent properties and this is something that although it has been I think very I think an activism and ecological psychology and other embodied views of cognition have been largely successfully in shifting this discussion to this point I think in the uh in more traditional cognitive Sciences you still have the idea that well organism organisms perceived by by describing the way the environment the the things this is something that I think uh more traditional readings of predictive processing make uh make clear as in uh the idea that you're inferring about the the source of stimuli and not in uh and not and this is not I mean this is more easily amenable to the idea that the brain or the cognitive system knows that the the world is so and so the environment is so and so instead of saying that the cognitive system knows how to engage with the world and I believe the the the the point that an activists and uh ecological psychologists are trying to make is to and also working on the the predictive coding and the free energy principle Etc is to show that these Notions these Frameworks could be a minimal idea that the cognitive system is knowing how to engage it knows how to engage the environment in a way it doesn't need to be read in a handboard summary that the the Kindle system is inferring that the world is so so I think this is one way of explaining but also uh philosophically if you have a large at the beginning of the the last few decades uh a gigantic discussion about whether you know how could be reduced to know that and then again you have a background of this and other philosophical discussion about the rather know how could be treated as an independent notion from you know that and whether you could reduce one to the other and so forth so I think it doesn't stop there you cannot uh even though I can make a a claim for perceptual some other knowing how they're still the the background issue of whether know-how could be treated as and all that and this is something that I am also working on recently there's a lot to say there it's it's such an interesting idea especially with linguistic communication it's a little bit like the finger and the moon and talking about the moon but the finger is only where it is and we're not talking about the pointer we're talking about the reference and so to use a sociocognitive form of communication in in cultured language or in written academic discourse to peer through that or to reimagine that and see even what seems to be like the Pinnacle of know that and still see it as know how I guess the question there is how do we read papers yes of course yes I think it is uh it is a uh it's yeah the values it seems to have to be a very big problem because if we try to to treat these negative capacities as you mentioned as a matter of know-how it seems that we still miss something

about communication for instance and on the other hand I think and this is my Innovation but I think that we could in principle talk about how understanding what you just said and trying to respond to it is a matter of knowing how to behave linguistically but I think that even this movement to treat high levels of combinatorialists which made it a matter of know-how it still leaves something out I am at I mean personally I am and pleased with it I think at some level we've got to say that well that is another kind of approach that is needed here to explain how we communicate for instance in periodically not to say that it does not involve and now but it's not just that I think that at least more than practical knowledge is needed but maybe the key here is to think of practical knowledge just basic and necessary for other sorts of other other kinds of knowledge that they are needed to engage in this sort of Exchange maybe maybe the maybe this is one layer I'm sure that well uh more has to be said about this to give them a more informative answer but this is a at least my current position is that no how provides the basis of our engagement with the world and with others but that some juncture we have to to be able to develop other kinds of understanding to explain a terrorists to explain the complex sorts of uh exchanges we have in our daily lives such as reading papers as you said I I agree I think framing know that as the basis and then recovering know how is a challenge and it's also a challenge to begin with the basal cognitive processes as being enacted no house and to reassemble those know-how pieces into know that again it's not necessarily even a simple challenge but would we rather confront the challenge of assembling symbolic propositional declarative verbal Etc cognitive processes from a more liminal ecological cognitive approach or try to recover the ecological from the turing machine in a sense so it seems like you're speaking in support of grounding in ecological plausible basal cognitive processes and then working towards assembling the skyscraper yes yes yeah totally yeah I think it's both are uh those both approaches are difficult to to to assemble as you said to to to provide this uh sort of explanation either bottom left or top down but I am more favorable of course to the the one web approach I think it also for another reason it it seems to the point of departure seems to be uh on biological description of how we interact with the world and the other way around it seems to be much more difficult to explain uh how we come up with this uh biological questions such as our body morphology and such it seems that we have to first account for that and then we we get to the the other questions which are also hard to to answer but at the very least I I think it's a the bottom up approach is more integrative you know is that the chances are it seems to be more uh in our favor in the sense and and the way you just kind of framed it made me think that if you were to program a computer from scratch it could be an abstract or a

virtual computer so it doesn't even need to have a specific embodiment and in that situation one might be concerned with assembling know that assert that and assembling these declarative statements and logic processes into something that is more dynamical or in interacting with the environment however that process of like full stack engineering with logic and silicone at the base is quite different from how we got here and so it might but it's it's complimentary um well you gave a little visibility into uh I guess the before during and after of peer review so I was just curious what changed during and after the paper like over the last months or years or however long it was how did you evolve on this and then what do you see as relevant next steps whether they're for you or just general to be done are we looking for a brain measurement are we looking for a video camera in a forest are we looking for a conceptual Advance what is going to move forward past your developments that's that's a great question I think I I'm not completely satisfied with the way I always find the people at the conceptual level I don't know if this any of these would be carried out in uh I honestly don't know if you could in principle carried out in other empirical terms but uh I'm not saying that uh well one is I would write it a bit differently today not that it's uh not that I disagree with the paper originally but I think I there's something some points that I would like to reframe but they do not authorize I think he had a new paper but maybe in the future but this is uh the questions such as the myths of the evening which is still the philosophical question and conceptual one they still uh bother me but I have yet to find opportunity to develop these ideas more carefully and the the general view I put forth there I think I I think it still stands uh but I would like I I still I'm waiting for this chance to elaborate it further and see how far I can get but I don't know yet I must admit I don't know what are the the future developments of this idea but um I've been working on several isolated issues on this approach for the last few years I've worked with the notion of information and scientific understanding and information for ecological and and activists of combination and recently I am working more recently with a few colleges on virtual reality and perception recognition so uh this is how I like to to conduct my research to find interesting uh questions and see how how can I deal with them but not necessarily find myself uh tied to one one line of reasoning by the way so this is what happened at the time I was working with rationality if I thought it was a very cool subject and uh I still think it is but I haven't had the time to come back and reframe what I wanted to but uh maybe maybe sometime in the near future I don't know but more the most recent thing I've been working on does not involved in cellular rationality but I think some products could be wrong which is that of uh virtual reality and how does uh visual reality actually compares to

combination because well there's obviously something missing from a perseverance in a visual environment right um there is of course if I someone for instance if I'm on a visual environment where I am under a kind of threat I may feel something analogous to fear but I'm not really fearing for my life and imagine this that the person on the virtual reality so is it rational to to run away from the the that threat the visual threat and so on so forth this is this the the this is what I'm working on and uh maybe it's it provides a point of return to this the idea of radical and active rationality and in fact uh the paper just got accepted so as I was telling you before I was so busy these last few days because we were correcting proofs my colleagues and I never figured uh and I I was with a tight schedule too to submit the corrected version of the paper in the last few weeks and then I got the proofs and well maybe uh in the future I can come back to the issue of rationality I'd like to but I don't know if I I don't know I I really can't foresee the developments of that but uh in general I still agree with my original idea but I'll frame it differently a little bit so the VR point I'm sure that there's a lot more to say there but it reminded me of just the title virtual reality it's a virtualization and so in the same way cognitive action planning allows us to entertain policies that our body doesn't take if only to evaluate them and choose against them but we can go on a flight of fancy have a Whimsical policy selection process of counterfactuals mentally but then our morphology is kind of like our Lifeline you know if you actually fall off a ledge then it's going to hurt and then in virtual reality you're embodied perceptual experience is quite literally a virtualization of survival critical situations so you might see yourself as being on a ledge but you're on a flat floor and then in that situation if you're kind of um a cognitive Observer outside of the video game you might say well it's rational for me just to fall off to see what happens so that's like but then there's a layer of rationality within the space yes like what's a good move in the game um and so it's kind of blurring the line between it's it's blurring the criticality of the body and um I'm sure you've thought about this much more so how does this play into our virtual reality um with the ways we think about and use and design these systems well this is the the the the the the questions that motivated us to write a paper and uh not uh give them and I was that in the technical literature you see that a lot of researchers working with the uh treat uh visual perception as illusory a kind of Illusion but we were a bit bothered by that because on the other hand it only works in VR because we we embodied the devices devices they work for us you have the controllers you have the the headset and uh it seems so weird to say that the user the the user is under a kind of Illusion and on the other hand it seems that it's not actually cognition because as you said there is no there is no biological relevance to those uh cognitive

States so to speak for instance if I see an apple in VR it's it does not afford nourishing but it makes sense within the game right but the different uh well what I perceive an apple in a in an actual environment it is it affords nourishment it is something I can eat but not in VR I can pretend to so we coined this concept of allusion to explain it is as if we were facing an actual animal but it involves also this volition of play this idea that we are not being sled because we put on the equipment we know we are in the game and this I think is a hot subject right now because there's metaphors method and I am very skeptical of whatever that can people are very optimistic but about the possibilities of the metaverse and some others really think that we could be in a simulation institutely speaking but I mean the best technology the current technology shows us is that it's very limited and you can pretend to be in a certain environment and so forth and this I think may offer a new new ways of thinking of thinking about rationality in this visual environment in ways that maybe Russians do something that environment has said to jump off the television see what happens and on the other hand behavioral aspects of doing so sometimes people actually feel fear and people face the street for phobias with virtual realities um it's it's a lot of there's a lot of interesting questions there that maybe I mean for future research we can yeah back to it it made me think about how the Apple in real life it could be calories for you but also it could be calories for a fly it could be thrown you could use it as a hammer you could paint it right so ecology and the generative process as we'd say an active inference kind of is always going Beyond whatever affordances were programmed in for a given object and then also um the current discourse on VR focuses heavily on headsets just speaking totally anecdotally not not my expert area um it seems a little bit interesting it's like as if the mobile phone discussion was only focused on rectangles of a certain size what makes this not virtual reality it's like looking through a window and there's another fellow to speak with so it's not attached to my head it doesn't have some of the motion tracking um capacities that some technologies have but it is a niche modification in Perry personal space and we can push signals and affordances to each other so having a VR framing or really just a framing of rationality and perception cognition action Cycles aren't just constrained to one specific realization or Hardware type and could run the gamut from Total offline person in a field to somebody who's very in The Matrix seems pretty relevant otherwise new technologies even little variants and software updates will continue to just infinitely surprise and confuse yeah I think it's a it's a it's hard to define the uh I mean we are here in an exchange which I think it could in principle be classified as a form of virtual reality of course I'm seeing your own your C mind but uh I mean most I think works out they

nowadays they they find it a doc by saying that well must have a headset and sometimes a controller but but other than that it's not spiritual reality but yeah it's hard to Define anything uh it's not it's not a dump and straightforward as one will think because on uh talking commercially of course virtual reality means headsets and and other paraphernalia but a phone just talking on the phone or a telegraph in that time it would have been on that interface with that sensory modality and with those affordances it is the virtualization yeah that we have access to of just a different chapter today yeah I think so yes part of our paper was to argue that in a way embodying technological devices is obviously in the need not that's not new it is the mark of Being Human it's embodying technological devices what made us what we are since the beginning using analytic technology and the fact that we can embody it so proficiently a device even if it is programmed to for instance uh in within virtual reality to have something a creature you control us richer with new morphology another morphology for instance even if that's the case it only happens so proficiently because we are able to embody the device some experiments have shown that you can map for instance a Twist of your wrist to a third arm uh that individual reality goes from was that from your chest and people can learn to control this third arm that doesn't map into our actual morphology but it can be made so by a Twist of the wrist so this is something we will discussed in our paper that well you can you can only do that because you embody the device so you can see it's an illusion it's actually well part of what means being human is to embody technological devices so I would go that way I mean I know it blurs the distinction between virtual and non-virtual but I think that's one way of dealing with this cool is there anything else you want to talk about ask or just kind of Prelude ah well I I'd like to thank you for the opportunity to discuss this paper it was very nice for me I I really enjoyed it and this is uh very nice and uh comfortable environment to discuss these issues and think I think once again thank you yeah awesome discussion and important topics and we'll be happy to talk to you again anytime okay thank you farewell thank you